

Non-parametric regression II – Handout

1 Exercise 2

Let us consider a univariate nonparametric model. As shown during the lesson (*c.f.* slide 271), the basis functions allows the linear representation:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \epsilon.$$

In order to control the wiggleness, the parameters $\boldsymbol{\beta}$ are estimated *via* minimization of a penalized cost function:

$$\mathcal{C}(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda\boldsymbol{\beta}^T \mathbf{S}\boldsymbol{\beta}.$$

In this exercise, we reparametrize the model in order to better understand some of its features. The so-called *natural parametrization* is described in the course (slide 292).

- a) Following the course, define $\mathbf{P} = \mathbf{U}^T \mathbf{R}$, $\boldsymbol{\beta}' = \mathbf{P}\boldsymbol{\beta}$ and $\mathbf{X}' = \mathbf{Q}\mathbf{U}$ (allowing $\mathbf{X} = \mathbf{X}'\mathbf{P}$), and express the cost function $\mathcal{C}(\boldsymbol{\beta}')$ under this new specification.

$$\begin{aligned} \mathcal{C}(\boldsymbol{\beta}') &= \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \lambda\boldsymbol{\beta}^T \mathbf{S}\boldsymbol{\beta} = \|\mathbf{y} - \underbrace{\mathbf{Q}\mathbf{U}}_{\mathbf{X}'} \underbrace{\mathbf{U}^T \mathbf{R}\boldsymbol{\beta}}_{\boldsymbol{\beta}'}\|_2^2 + \lambda\boldsymbol{\beta}^T \mathbf{R}^T \underbrace{\mathbf{R}^{-T} \mathbf{S} \mathbf{R}^{-1}}_{\mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^T} \mathbf{R}\boldsymbol{\beta} \\ &= \|\mathbf{y} - \mathbf{X}'\boldsymbol{\beta}'\|_2^2 + \lambda \underbrace{\boldsymbol{\beta}'^T \mathbf{R}^T \mathbf{U}}_{\boldsymbol{\beta}'^T} \underbrace{\mathbf{U}^T \mathbf{R}\boldsymbol{\beta}}_{\boldsymbol{\beta}'} = \|\mathbf{y} - \mathbf{X}'\boldsymbol{\beta}'\|_2^2 + \lambda\boldsymbol{\beta}'^T \boldsymbol{\Lambda}\boldsymbol{\beta}' \end{aligned}$$

- b) Provide the estimator $\hat{\boldsymbol{\beta}}'$ of $\boldsymbol{\beta}'$ minimizing $\mathcal{C}(\boldsymbol{\beta}')$.

$$\begin{aligned} \mathcal{C}(\boldsymbol{\beta}') &= \|\mathbf{y} - \mathbf{X}'\boldsymbol{\beta}'\|_2^2 + \lambda\boldsymbol{\beta}'^T \boldsymbol{\Lambda}\boldsymbol{\beta}' = (\mathbf{y} - \mathbf{X}'\boldsymbol{\beta}')^T (\mathbf{y} - \mathbf{X}'\boldsymbol{\beta}') + \lambda\boldsymbol{\beta}'^T \boldsymbol{\Lambda}\boldsymbol{\beta}' \\ &= \mathbf{y}^T \mathbf{y} - \boldsymbol{\beta}'^T \mathbf{X}'^T \mathbf{y} - \mathbf{y}^T \mathbf{X}' \boldsymbol{\beta}' + \boldsymbol{\beta}'^T \mathbf{X}'^T \mathbf{X}' \boldsymbol{\beta}' + \lambda\boldsymbol{\beta}'^T \boldsymbol{\Lambda}\boldsymbol{\beta}' \\ &= \mathbf{y}^T \mathbf{y} - \boldsymbol{\beta}'^T \mathbf{X}'^T \mathbf{y} - \mathbf{y}^T \mathbf{X}' \boldsymbol{\beta}' + \boldsymbol{\beta}'^T [\mathbf{X}'^T \mathbf{X}' + \lambda\boldsymbol{\Lambda}] \boldsymbol{\beta}' \\ \frac{\partial \mathcal{C}}{\partial \boldsymbol{\beta}'}(\boldsymbol{\beta}') &= -2\mathbf{X}'^T \mathbf{y} + 2[\mathbf{X}'^T \mathbf{X}' + \lambda\boldsymbol{\Lambda}] \boldsymbol{\beta}' \end{aligned}$$

Finally :

$$\frac{\partial \mathcal{C}}{\partial \beta'} (\hat{\beta}') = 0 \Leftrightarrow -2\mathbf{X}'^T \mathbf{y} + 2 [\mathbf{X}'^T \mathbf{X}' + \lambda \mathbf{\Lambda}] \hat{\beta}' = 0 \Leftrightarrow \hat{\beta}' = [\mathbf{X}'^T \mathbf{X}' + \lambda \mathbf{\Lambda}]^{-1} \mathbf{X}'^T \mathbf{y}$$

- c) Deduce the linear smoother $\mathbf{A}$ such that $\hat{y} = \mathbf{A}y$. Compute $\text{tr}(\mathbf{A})$ and find an interpretation for this quantity.

Thus, this penalized regression smoother provides the following *fitted values*:

$$\hat{y} = \mathbf{X}' \hat{\beta}' = \underbrace{\mathbf{X}' [\mathbf{I} + \lambda \mathbf{\Lambda}]^{-1} \mathbf{X}'^T}_{\mathbf{A}} \mathbf{y} = \mathbf{A}y$$

The matrix $\mathbf{A}$ is thus a projection (or *hat*) matrix with rank equal to its trace. Thus, the *effective degrees of freedom* of the smoothing procedure can be computed via the computation of this trace.

$$\begin{aligned} \text{tr}(\mathbf{A}) &= \text{tr}(\mathbf{X}' [\mathbf{I} + \lambda \mathbf{\Lambda}]^{-1} \mathbf{X}'^T) = \text{tr}(\mathbf{X}'^T \mathbf{X}' [\mathbf{I} + \lambda \mathbf{\Lambda}]^{-1}) = \text{tr}([\mathbf{I} + \lambda \mathbf{\Lambda}]^{-1}) \\ \text{tr}(\mathbf{A}) &= \sum_{k=0}^K \frac{1}{1 + \lambda \Lambda_{k+1,k+1}} < K + 1 \end{aligned}$$

This value is smaller than $K + 1$, since the values in the denominator are larger than 1.

- d) Explain what happens to $\mathbf{A}$ as λ increases.

As λ increases, we reduce the effective degrees of freedom in the smooth. When λ is close to zero, the effective degrees of freedom are equal to $K + 1$, meaning that all parameters have been used to represent the smooth. On the other hand, it is not excluded that this sum adds up to 1, implying that a linear fit would be sufficient to model the impact of the covariate on the response.

The interpretation of the *edf* extrapolates to the estimates obtained with Generalized Additive Models with Smoothing Parameters obtained by minimizing a criterion (UBRE, GCV, etc.).